NOTICE 1
23 October 2012

TEST OPERATIONS PROCEDURE

This Test Operations Procedure (TOP) has been archived per ATEC memorandum dated 18 October 2012.

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UNCLASSIFIED
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DOCUMENT CONTROL DATA - R & D

Security classification of title, body of abstract and indexing annotation must be entered when the overall report is classified)

1. ORIGINATING ACTIVITY (Corporate author)
US Army Test and Evaluation Command (USATECOM)
Aberdeen Proving Ground, Maryland 21005

2a. REPORT SECURITY CLASSIFICATION
Unclassified

2b. GROUP
---

3. REPORT TITLE
US Army Test and Evaluation Command Materiel Test Procedure 8-2-072, Commodity Engineering Test Procedure, - "Sampling and Analyzing Kits, CBR Agent."

4. DESCRIPTIVE NOTES (Type of report and inclusive dates)
Final

5. AUTHOR(S) (First name, middle initial, last name)
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6. REPORT DATE	7a. TOTAL NO. OF PAGES	7b. NO. OF REFS
3 March 1970	26	12

8a. CONTRACT OR GRANT NO.	9a. ORIGINATOR'S REPORT NUMBER(S)
DA-18-001-AMC-1045(R)	
b. PROJECT NO AMCR 310-6	MTP 8-2-072
c.	9b. OTHER REPORT NO(S) (Any other numbers that may be assigned this report) ---
d.	

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11. SUPPLEMENTARY NOTES	12. SPONSORING MILITARY ACTIVITY
---	Headquarters US Army Test and Evaluation Command Aberdeen Proving Ground, Maryland 21005

13. ABSTRACT
This Engineering Test Procedure describes test methods and techniques for evaluating the technical performance and characteristics of Sampling and Analyzing Kits (CBR Agent Type). The evaluation is related to criteria expressed in applicable Qualitative Materiel Requirements (QMR), Small Development Requirements (SDR), Technical Characteristics (TC), or other appropriate design requirements and specifications.

DD FORM 1 NOV 65 1473 (PAGE 1)
S/N 0101-807-6811

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A-31408

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14 KEY WORDS	LINK A		LINK B		LINK C	
	ROLE	WT	ROLE	WT	ROLE	WT
Engineering Test						
Sampling and Analyzing Kits (CBR Agent Type)						
Chemical Biological Radiological (CBR)						
Test Procedures						
Test Methods and Techniques						

DD FORM 1 NOV 65 1473 (BACK)

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REFERENCE FILE

3 March 1970

U. S. ARMY TEST AND EVALUATION COMMAND
COMMODITY ENGINEERING TEST PROCEDURE

SAMPLING AND ANALYZING KITS, CBR AGENT

1. OBJECTIVE

The objective of this Materiel Test Procedure (MTP) is to determine the technical performance and safety aspects of CBR agent sampling and analyzing kits with respect to the criteria cited in applicable Qualitative Materiel Requirements (QMR), Small Development Requirements (SDR), Technical Characteristics (TC), and other requirements and documentation that pertain to the test item.

2. BACKGROUND

This MTP covers the testing of sampling and analyzing kits for CBR agents. These consist of a collection of items for use in detecting, sampling, and identifying toxic chemical agents and toxic smokes in the field, involving partial qualitative analysis, and in sampling biological warfare agents and radiological warfare agents for further processing by appropriate laboratories. The kits are used by trained military personnel and are packaged in portable carrying and storage cases.

3. REQUIRED EQUIPMENT

a. Suitable site for test operations
b. Environmental chambers for:

   1) Temperature testing
   2) Sunshine testing
   3) Water immersion testing
   4) Humidity testing
   5) Fungus testing
   6) Salt fog testing
   7) Dust testing

c. Photographic Equipment:

   1) Still camera
   2) Motion picture camera

d. Meteorological equipment to measure and record:

   1) Temperature
   2) Wind direction and velocity
   3) Relative humidity

e. Protective Clothing
f. Decontamination Equipment
g. Aircraft or Simulated Aircraft

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h. Accelerometers, as required
i. Transportability testing area for:

1) Vibration
2) Shock

j. Materials Handling Equipment
k. CBR Samples, as required

4. REFERENCES

A. USATECOM Regulation 385-6, Verification of Safety of Materiel During Testing.
B. MIL-STD-810B, Environmental Test Methods.
C. MIL-STD-1472, Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
D. MIL-H-46855, Human Engineering Requirements for Military Systems, Equipment and Facilities.
E. USAMC Pamphlet 706-134, Engineering Design Handbook: Maintainability Guide for Design.
F. Woodson, W.E., and D.W. Conover, Human Engineering Guide for Equipment Designers. Second Edition. Berkeley: University of California Press. 1966.
G. MTP 7-1-602, Airportability and Airdrop Service Testing.
H. MTP 7-2-509, Air Drop Capability of Materiel (General).
I. MTP 7-2-515, Air Transport (Suitability of Equipment for).
J. MTP 8-2-500, Receipt Inspection.
K. MTP 8-2-503, Rough Handling and Surface Transport.
L. MTP 8-2-510, Decontamination.

5. SCOPE

5.1 SUMMARY

This MTP describes in general terms the preparation, conduct, recording, and reporting methods for testing of sampling and analyzing kits which are used for the detection of chemical agents and for sampling chemical, biological, and radiological warfare agents.

The following subtests will be performed on a selective basis as required to determine if the test item meets the criteria established:

a. Receipt Inspection - An inspection of the test item as received to (1) determine its physical characteristics and condition, (2) locate any defects it might have, and (3) identify damage received during transport. During this inspection, test items will also be numbered serially for subsequent identification purposes.

b. Safety Evaluation - The objective of this evaluation is to check the safety statement issued by the developing agency, to insure that adequate safety features have been incorporated, and to obtain data to be included in the safety release recommendation required by USATECOM Regulation 385-6.

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c. Simulated Environmental testing - An evaluation to determine the effects of cyclic storage, extreme temperatures, humidity, fungus, dust, and fresh and salt water on the test item.

d. Rough Handling and Surface Transport tests - An evaluation to determine the effects of rough handling and surface transport on the physical and operational characteristics of the test item.

e. Air Transportability - An evaluation to determine the effects of subjecting the test item to air transport conditions.

f. Air Drop Capability - An evaluation to determine the effects on the test item resulting from its being subjected to air drop conditions. (delivery by parachute) and to identify the problems involved in delivery of the test item by parachute.

g. Decontamination Aspects - An evaluation to determine the relative ease or difficulty involved in decontamination of the test item, the effectiveness of decontamination, and its effect on test items.

h. Maintenance Aspects - An evaluation to determine if the test item design permits easy maintenance and if the instructions are adequate.

i. Operational Characteristics - An evaluation to determine (1) the sensitivity of the test item to specified agents, (2) its capability to provide the required response within the specified time limits, (3) any undesirable sensitivity to other agents and substances, and (4) reliability testing.

j. Field Operability - An evaluation to determine the capability of the test item to detect the specified agents under field conditions within the time limits and response requirements stipulated by governing criteria.

k. Human Factors - The objective of this subtest is to determine the characteristics of the test item that involve human factors considerations in handling and operating the test item.

5.2 LIMITATIONS

None.

6. PROCEDURES

6.1 PREPARATION FOR TEST

6.1.1 Safety Statement Verification

The test officer will verify that a Safety Statement has been received from the developing agency before testing is started.

NOTE: The Safety Statement shall include information about the operational limitations and potential safety hazards which are peculiar to the test item.

6.1.2 Personnel Safety

a. The test plan shall insure that tests are performed in the safest manner consistent with the accomplishment of the mission. The main intent of the plan shall be to limit the exposure to a minimum number of personnel, for a minimum amount of time, to a minimum amount of hazardous material consistent

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with safe and efficient operation. Plans shall include safety procedures, precautions and emergency procedures, as necessary. Information based on the test item Safety Statement shall be incorporated into the test plan. The evaluation of potential hazards; analysis of risks, limitations, special precautions, equipment and techniques shall be incorporated.

b. One individual will be responsible for the safety aspects of each test. He must understand the operation of the test items and the hazards of the test agents. He will review test procedures for the purpose of evaluating hazards and will recommend control measures.

c. All personnel who participate in or observe the test will be informed of the hazards and the precautions required as well as details of the test procedure to insure safety of the personnel.

6.1.3 Security

Security considerations will be adequately determined and properly applied to each situation.

6.1.4 Logistical Requirements

The test officer will insure that all logistical requirements are satisfied.

6.2 TEST CONDUCT

6.2.1 Receipt Inspection

The item under test shall be subjected to the applicable portions of MTP 8-2-500, and the following:

a. Examine the carrying or storage case for loose, broken, or damaged hardware and fittings, and rivets, if applicable.

b. Inspect the gasket material used to seal the carrying case for wear or deterioration.

c. Measure and record the dimensions and weight of the test item's carrying case.

d. Open units and check sampling kits for:

1) Breakage
2) Chemical Spillage
3) Chemical and material deterioration
4) Proper identification of all reagents
5) Expiration dates of components, if applicable

e. Check the test kit items against an approved contents list. Verify that the kit contains all of the required components.

f. Any damaged or discrepant items shall be set aside for either repair or replacement.

6.2.2 Safety Evaluation

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Determine the test item's safety as follows:

a. Observe the operation of the test item in accordance with existing instructions, instruction manuals, directives, safety SOP's, and similar guidance. Record any features of the test item which might constitute a hazard to personnel.

b. Throughout the conduct of other tests, hazardous features will be specifically observed and noted.

c. The safety aspects included in the Safety Statement prepared by the developing agency will be verified.

d. The data to be included in the Safety Release Recommendation required by Reference 4A (USATECOM Regulation 385-6) will be obtained.

6.2.3 Simulated Environmental Testing

6.2.3.1 Cyclic Storage

a. Subject the test items and their containers to cycles of climatic extremes. A cycle shall consist of three weeks duration as follows: successive one-week tests at humid, low temperature, and high temperature. Chamber conditions for each climatic conditions are as follows:

1) Humid Storage. The chamber shall be maintained at 113°F ± 2°F and 85% relative humidity for the duration of the test.
2) Low Temperature. The chamber shall be maintained at -65°F ± 2°F for the duration of the test.
3) High Temperature. The chamber shall be maintained at 160°F ± 2°F for the duration of the test.

b. The test items shall be subjected to a minimum of three such cycles, or more, if specified. Upon completion of each cycle the container and its contents shall be examined for damage.

6.2.3.2 Extreme-Temperature Tests

Unless otherwise directed the test item will be subjected to the following temperature tests:

6.2.3.2.1 Low Temperature Tests - Place a minimum of two test items in a test chamber, and perform the following:

a. Reduce the chamber temperature to the lowest temperature at which it may be stored, maintain it at this temperature for a period of 72 hours.

b. Visually inspect the test items in the chamber and record any damage.

c. Adjust the chamber temperature to the test item's minimum operating temperature as established by design requirements, and maintain this temperature until stabilization has been reached. If stabilization is attained in less than 24 hours, maintain this temperature for a complete 24 hour period

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and perform the following:

NOTE: Stabilization is considered to be reached when the temperature of the test item does not change more than 2°C (3.6°F) per hour.

1) Visually inspect the test items and report any damage.
2) Remove 1/2 of the test items and verify their operability by subjecting them to the applicable procedures as described in paragraph 6.2.9.

NOTE: Operability checks should be accomplished within 15 minutes of removing the test items from the chamber.

d. Remove the items from the chamber, allow their temperature to stabilize at local ambient conditions, then perform the following:

1) Visually inspect the test items, record any discrepancies.
2) Verify the operability of the remaining items by subjecting them to procedures listed in paragraph 6.2.9.

6.2.3.2.2 High Temperature Tests - Place a minimum of two test items into the temperature chamber, and perform the following:

a. Adjust the temperature of the chamber to the highest temperature at which the test item is designed to operate, and adjust the chamber humidity to 15% relative. Maintain these conditions for a minimum of 24 hours, then perform the following:

1) Visually inspect the items under test and record any damage.
2) Without removing the test items from the chamber, verify the operability of 1/2 of the test items by subjecting them to the procedures of paragraph 6.2.9.

b. Remove all of the remaining test items from the chamber and allow them to stabilize at local ambient temperature and humidity for 24 hours, and perform the following:

1) Visually inspect the test items and record any damage.
2) Verify the operability of the test items by subjecting the remaining items to the procedures in paragraph 6.2.9.

6.2.3.3 Fungus Test

a. Subject a minimum of two test items to the fungus exposure conditions of Procedure I, Method 508, Reference 4B (MIL-STD-810B).

b. At the completion of the exposure period, perform the following:

1) Visually inspect the items and record any evidence of fungus, or corrosion.
2) Disassemble 1/2 of the test items and inspect the components

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for signs of fungus or corrosion, if applicable.

3) Verify the operability of the item by subjecting the remaining items to the applicable procedures of paragraph 6.2.9.

6.2.3.4 Humidity Test

a. Subject a minimum of 2 test items to the humidity cycling of Procedure I, Method 507, Reference 4B (MIL-STD-810B).

b. At the completion of the cycling period, perform the following:

1) Visually inspect the items and record any signs of corrosion.
2) Disassemble 1/2 of the test items, and check for corrosion.
3) Verify the operability of the remaining test items by subjecting them to the procedures listed in paragraph 6.2.9.

6.2.3.5 Dust Test

a. Subject a minimum of two test items to exposure conditions of Procedure I, Method 510, Reference 4B (MIL-STD-810B).

b. At the completion of the exposure period, perform the following:

1) Visually inspect the test items and record any surface damage.
2) Examine the interior of the test item cases and the contents for damage and dust.
3) Disassemble 1/2 of the test items, and inspect the components for damage and presence of dust.
4) Verify the operability of the test items by subjecting them to the applicable procedures of paragraph 6.2.9.

6.2.3.6 Sunshine Test

a. Subject a minimum of two test items to the sunshine conditions of Procedure I, Method 505, of Reference 4B (MIL-STD-810B).

b. At the completion of the exposure period, perform the following:

1) Visually inspect the test items and their components, and record any damage and deterioration noted.
2) Disassemble 1/2 of the test items and inspect the components for evidence of deterioration.
3) Subject the remaining test items to the applicable procedures of paragraph 6.2.9.

6.2.3.7 Water Immersion Tests

a. Subject a minimum of two test items to the water immersion conditions of Procedure I, Method 512, Reference 4B (MIL-STD-810B).

NOTE: If design requirements establish depth of water, water temperature, or time of immersion different from the standard procedure, the test plan will so state.

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b. At the completion of the test, perform the following:

1) Open the test item and inspect the contents for evidence of water penetration.
2) Verify the operability of the test items by subjecting them to the applicable procedures of paragraph 6.2.9.

6.2.3.8 Salt Fog Test

a. Subject a minimum of two test items to the conditions of Procedure I, Method 509, Reference 4B (MIL-STD-810B).

b. At the completion of the salt fog spray exposure, perform the following:

1) Rinse the test item cases with clear water.
2) Visually inspect the test items and their contents, and record all evidence of water penetration and corrosion.
3) Disassemble 1/2 of the test items and inspect their components for evidence of water penetration, and corrosion.
4) Verify the operability of the test items by subjecting the remaining test items to the procedures of paragraph 6.2.9.

6.2.4 Rough Handling and Surface Transport Test

a. Subject a minimum of two test items, crated or packaged for shipment, to the following procedures of MTP 8-2-503:

1) Procedure 6.2.2.2a.3
2) Procedure 6.2.2.1a.2

b. At the completion of the test, perform the following:

1) Visually examine the test items and record any evidence of breakage, leakage, or other damage to the contents.
2) Disassemble 1/2 of the items tested and inspect the components for damage.
3) Verify the operability of the remaining test items using the procedures established in paragraph 6.2.9.

6.2.5 Air Transportability

Determine the effects of altitude and vibration, similar to that which will be experienced by the test item in flight and in handling during loading and unloading operations, by performing the following:

NOTE: Background information on air transportability is contained in MTP 7-1-002.

6.2.5.1 Loading/Unloading

Determine the ease of loading and unloading the aircraft as described

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in the applicable portions of MTP 7-2-515, or as follows:

a. Load the test items (in their shipping containers, if applicable) aboard aircraft or simulated aircraft as indicated in the test plan loading schedule using normal loading equipment and record the following:

1) Type of aircraft used/simulated
2) Shipping container dimensions and weight
3) Equipment used for loading
4) Method of tiedown
5) Damage to package occurring during loading

b. Unload the test items from the aircraft/simulated aircraft and record:

1) Equipment used in unloading
2) Difficulties encountered during unloading operations
3) Damage sustained by the shipping container during unloading

6.2.5.2 Simulated Flight Test

a. Subject a minimum of two test items (in shipping containers, if applicable) to the following simulated conditions simultaneously:

1) Pressure equal to that of the maximum altitude that the test item is expected to be flown.
2) Flight vibration conditions as described in Procedure I, Method 514, of Reference 4B (MIL-STD-810B).

b. At the completion of this test repeat the procedures of paragraph 6.2.4, step b.

6.2.6 Air Drop Capability

The air drop characteristics of the test item, when in its shipping containers will be determined as described in the applicable sections of MTP 7-2-509, and as follows:

6.2.6.1 Free Fall Test

NOTE: Perform the following prior to conducting the procedures of paragraph 6.2.6.2:

Subject a minimum of 4 test items, packaged in their original containers, to a free fall drop test.

a. Using a crane with a quick release hook, raise the test item, over a specified type surface, to the height specified in the test plan.
b. Release the test item and record the following:

1) Type of surface

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2) Height of drop
3) Acceleration magnitude at impact

c. Document the drop test by both still and motion picture cameras.
d. At the completion of the test, perform the following:

1) Visually examine the test items package, and record the presence of cracks, breaks, undone bindings, etc.
2) Visually examine the test items for, and record the presence of, damage.
3) Verify the operability of the test items by subjecting them to the procedures of paragraph 6.2.9.

6.2.6.2 Air Drop by Aircraft

Determine the airdrop characteristics of the test item, when in its shipping containers, as described in the applicable sections of MTP 7-2-509, and as follows:

a. Rig a minimum of 4 test items in the appropriate air drop containers and drop the containers from aircraft flying at the altitude and speed stipulated in the test plan. Record the following:

1) Aircraft used
2) Aircraft altitude
3) Aircraft airspeed
4) Meteorological conditions
5) Air delivery system trajectory and impact velocities
6) Acceleration magnitude at impact, in g's

b. Conduct coverage of the air drop test procedures with motion and still cameras.
c. At the completion of the test, perform the following:

1) Visually examine the test item's package, and record the presence of cracks, breaks, undone bindings, etc.
2) Visually examine the test item for and record the presence of damage.
3) Verify the operability of the test items by subjecting them to the procedures of paragraph 6.2.9.

6.2.7 Decontamination Aspects

a. Decontaminate the test item as described in the applicable portions of MTP 8-2-510.
b. Record any difficulties encountered.
c. Determine and record the effectiveness of each method of decontamination used.
d. Verify the operability of the test item after decontamination as described in paragraph 6.2.9.

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6.2.8 Maintenance Aspects

a. Determine the test item maintenance aspects using the criteria of USAMC Pamphlet 706-134 as a guide.

NOTE: The features of design which permit, enhance, or hinder the accomplishment of maintenance by personnel of average skills under environmental conditions similar to those in which maintenance is to be performed will be recorded.

b. Determine and record the following as required:

1) Ease of performing maintenance
2) Component interchangeability
3) Adequacy and accuracy of the maintenance documentation
4) Adequacy of spare parts supply
5) Maintenance category of the test item, if specified
6) Mean time-to-repair
7) Special tools required
8) Number of personnel required and special skills
9) Recommendations pertaining to improvements that could be made

c. Use photographs as appropriate to document maintenance procedures.

6.2.9 Operational Characteristics

These tests will be conducted so that the capability of the test item to meet minimum time, reliability, and sensitivity requirements, as set forth in the applicable materiel requirements documents, can be determined. The standard item being replaced should be tested concurrently for comparative evaluation.

6.2.9.1 Sensitivity and Response

a. Determine with what reliability the test item responds to the minimum concentration in the maximum response time, and record the threshold concentration detection limits for chemical agents in or on:

1) Air
2) Water
3) Surfaces

NOTE: 1. Prescribed concentrations of designated chemical agent vapor will be provided by an appropriate gas generator. Sampling will be performed using the test item.
2. The number of determinations to be made for each chemical agent will be specified in the test plan.

b. Repeat the determinations at prescribed intervals throughout the entire test cycle to insure that the sensitivity of the test item does not become degraded with operation, time, or specific types of testing.

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NOTE: Consistency and reliability of responses will be verified by repetitive testing to establish a pattern or response characteristics and limitations of responses.

c. Repeat steps a and b for radioactive sources using procedures as indicated in the manufacturer's instructions accompanying the equipment.

d. Repeat steps a and b for biological samples in accordance with the manufacturer's instructions accompanying the equipment and other current technical manuals.

6.2.9.2 Interference Aspects

a. Determine and record the threshold concentration detection limits for CBR agents in the presence of various other contaminants in:

1) Water
2) Other liquids
3) Vapors
4) Smokes

NOTE: Agents and substances other than those the test item is designed to detect will be suitably dispersed so that the test item is exposed to them. These will normally include common substances such as those that might be encountered under actual conditions of use. The test item will be operated in the normal manner and interferences will be recorded.

Examples of interfering contaminants might include the following:

1) Signaling and screening smokes
2) Smoke from burning wood, rubber, rubbish
3) Fumes from decomposing animals and waste
4) Insecticides and defoliants
5) Ammonia
6) Explosion fumes
7) Fog oil smoke from smoke generator
8) Petroleum fire smoke
9) Engine exhaust gases
10) Decontaminants
11) Dust, fog, salt fog

b. Record the following:

1) Interference contaminant identity
2) CBR agent identity
3) Response times for various agent concentrations
4) False responses

6.2.10 Field Operability

a. The test item will be operated under field conditions in accordance with the applicable criteria for the item being tested.

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b. Meteorological conditions and limitations will be as prescribed in the applicable materiel requirements criteria. Tests will not normally be conducted in winds greater than 16 kilometers per hour nor during periods of significant precipitation.

c. All tests will be repeated for each agent as specified in the test plan.

d. Personnel participating in field trials will be adequately protected by protective clothing and equipment in accordance with safety requirements.

e. Color photographs will be taken as required to record typical color reactions of the test item where applicable.

f. False responses will be noted and possible or probable causes listed when known. The capability of the test item to respond to and detect prescribed agents in the field at the levels of interest will be noted.

6.2.10.1 Air Tests

Detect the presence of specified agents in air under the prescribed conditions, as follows:

a. Disseminate the agent by any suitable dissemination method.

b. As soon as possible after the dissemination, test personnel will move into the area of contamination and proceed to identify the agent using standard operating procedures. Several soldiers will carry test kits; others will carry standard pretested kits. Checks will have been made to ascertain that the contents of the standard kits have not deteriorated.

c. The personnel performing the test will not know for which agent they are testing. They will work independently within the area and will not compare the results. This test should be performed in the vicinity of quantitative samplers.

6.2.10.2 Water Tests

Detect the presence of specified agents in contaminated water as follows:

a. Trained personnel will test a reservoir or containers of water previously exposed within the agent dissemination area. A sufficient number of tests will be performed to provide a valid data basis for calculating a reliability estimate.

b. Test personnel will take a sample of the water in the plastic bags provided; however, glass vials should be used for biological samples. The water will be analyzed immediately in the laboratory for the known agent and for the agent reported by the personnel using the detection kits.

6.2.10.3 Surface Tests

Detect the presence of specified agents on contaminated surfaces as follows:

a. Trained test personnel will test for agent on sample plates of

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glass and metal previously exposed within the agent dissemination area. For biological contamination, standard sampling procedures will be used.

b. Test item performance will be compared with results obtained with control items.

c. If additional evaluation is required, perform laboratory procedure to obtain analysis for comparison purposes.

6.2.11 Human Factors

During conduct of all subtests observations will be made regarding the human factors engineering characteristics of the test item. (Consult Reference 4C (MIL-STD-1472) and Reference 4D (MIL-H-46855) for discussions of the human factors relevant to the items under test). Observe and record the following:

a. Carrying strap adequacy.
b. Ease of handling.
c. Ease of carrying.
d. Ease of operating.
e. Ease of interpretation of response by the item operator (level of skill required).
f. Ease of interpretation of responses under various lighting conditions.
g. Compatibility with other clothing and equipment.
h. Ease of handling items when wearing protective clothing, gloves, etc. (including cold-weather clothing).
i. Adequacy and simplicity of operating instructions.
j. Time required for interpreting response.
k. Factors which caused frequent complaints from operators.

6.3 TEST DATA

6.3.1 Receipt Inspection

a. Record the data collected as described in applicable sections of MTP 8-2-500 and the following:

1) For the carrying case:

a) Test item identification number and nomenclature
b) Loose, broken or damaged hardware or fittings
c) Evidence of wear or damage to gasket
d) Dimensions, in inches
e) Weight, in pounds

2) Evidence of the following for the test item components:

a) Breakage
b) Chemical leakage
c) Chemical and material deterioration
d) Identification of missing components

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3) For the chemicals:

a) Identification of each
b) Expiration date, if applicable

b. Retain all photographs.

6.3.2 Safety Evaluation

Record the following:

a. Any hazardous characteristics
b. Test item deficiencies and recommendations for improvement
c. Information for inclusion in the Safety Release Recommendation

6.3.3 Simulated Environmental Testing

6.3.3.1 Cyclic Storage

Record the following for each cycle:

a. Test item identification number
b. Cycle number
c. Damage to:

1) Container
2) Test item

6.3.3.2 Extreme Temperature Tests

6.3.3.2.1 Low Temperature Tests -

Record the following for each test item, as applicable:

a. Test item identification number
b. For minimum storage temperature:

1) Temperature in °C and °F
2) Damage incurred

c. For minimum operating temperature:

1) Temperature in °C and °F
2) Damage incurred
3) Operability data collected as described in paragraph 6.2.9

d. For ambient temperature:

1) Temperature in °C and °F
2) Test item damage
3) Operability data collected as described in paragraph 6.2.9

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6.3.3.2.2 High Temperature Tests -

Record the following for each test item, as applicable:

a. Test item identification number
b. The maximum operating temperature:

    1) Damage incurred
    2) Operability data collected as described in paragraph 6.2.9

c. For ambient temperature:

    1) Temperature in °C and °F
    2) Relative humidity in percent
    3) Damage incurred
    4) Operability data collected as described in paragraph 6.2.9

6.3.3.3 Fungus Test

Record the following for each test item:

a. Test item identification number
b. Operability data collected as described in paragraph 6.2.9
c. Presence of fungus on test item and components
d. Evidence of corrosion

6.3.3.4 Humidity Test

Record the following for each test item:

a. Test item identification number
b. Operability data collected as described in paragraph 6.2.9
c. Evidence of corrosion on the test item and components

6.3.3.5 Dust Test

Record the following for each test item:

a. Test item identification number
b. Operability data collected in paragraph 6.2.9
c. Damage to:

    1) External surfaces
    2) Test item components

d. Presence of dust on test item components

6.3.3.6 Sunshine Test

Record the following for each test item:

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a. Test item identification number
b. Surface damage
c. Deterioration of components
d. Operability data collected as described in paragraph 6.2.9

6.3.3.7 Water Immersion Test

Record the following for each test item as applicable:

a. Test item identification number
b. During immersion:

1) Presence of bubbling, if any
2) Immersion time to bubbling, if any, in minutes

c. Presence of water penetration
d. Operability data collected as described in paragraph 6.2.9

6.3.3.8 Salt Fog Test

Record the following for each test item as applicable:

a. Test item identification number
b. Operability data collected as described in paragraph 6.2.9
c. Evidence of corrosion on test item and components
d. Evidence of water penetration

6.3.4 Rough Handling and Surface Transport Test

Record the following for each test item, as applicable.

a. Test item identification number.
b. Data collected as described in the applicable sections of MTP 8-2-503.
c. Operability data collected as described in paragraph 6.2.9.

6.3.5 Air Transportability

6.3.5.1 Loading/Unloading

Record the following:

a. Data collected as described in the applicable sections of MTP 7-2-515.
b. Type of aircraft used or simulated
c. Shipping container:

1) Length, width and height, in inches
2) Weight, in pounds
3) Material

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d. Equipment used in loading
e. Difficulties encountered while loading
f. Method of tiedown
g. Damage incurred to the package while loading
h. Equipment used in unloading
i. Difficulties incurred in unloading
j. Damage incurred to the package while unloading

6.3.5.2 Simulated Flight Test

Record the following for each item under test:

a. Test item and identification number
b. Altitude simulated, in feet
c. Ambient pressure in test chamber, in inches of Hg
d. Data collected as described in paragraph 6.2.4.b

6.3.6 Air Drop Capability

6.3.6.1 Free Fall Test:

a. Record the following:

1) Type of impact surface
2) Height of drop, in feet
3) Acceleration magnitude at impact in g's
4) For the test item package:

a) Presence of cracks, breaks, etc.
b) Undone bindings

5) For the test item:

a) Damage or deformities
b) Operability data collected as described in paragraph 6.2.9

b. Retain all photographs.

6.3.6.2 Air Drop by Aircraft

a. Record the following for each test item:

1) Test item identification
2) Type aircraft used
3) Aircraft altitude, in feet
4) Aircraft airspeed, in mph
5) Air conditions (calm, turbulent)
6) For meteorological conditions:

a) Temperature in °C
b) Wind speed in mph

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c) Wind direction
d) Relative humidity in percent
e) Barometric pressure

7) Deceleration magnitude at impact in g's
8) Impact velocity in fps
9) For test item package:

a) Presence of cracks, breaks, etc.
b) Undone binding

10) For the test item:

a) Damage or deformities
b) Operability data collected as described in paragraph 6.2.9

b. Retain all motion and still pictures

6.3.7 Decontamination Aspects

Record the following for each test item undergoing decontamination:

a. Data collected as described in the applicable sections of MTP 8-2-510.

b. Operability data collected as described in the applicable procedures of paragraph 6.2.9.

6.3.8 Maintenance Characteristics

a. Record the following:

1) Ease of performing maintenance
2) Component interchangeability
3) Adequacy and accuracy of the maintenance documentation
4) Adequacy of spare part supply
5) Maintenance category of the test item, if specified
6) Mean time-to-repair
7) Special tools required
8) Number of personnel and special skills required
9) Recommendations pertaining to improvements that could be made

b. Retain all photographs

6.3.9 Operational Characteristics

6.3.9.1 Sensitivity and Response

Record the following for each determination:

a. Test item identification number
b. Threshold agent concentration (in applicable units) for:

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1) Air
2) Water
3) (on) Surfaces

c. Determination number (1, 2, 3, 4, 5)
d. Response time in seconds
e. Agent identity
f. Determination identity (air, water, etc.)

6.3.9.2 Interference Aspects

Record the following for each determination:

a. Test item identification number.
b. Threshold agent concentration (in applicable units) for:

1) Water
2) Other liquids
3) Vapors
4) Smokes

c. Determination number (1, 2, 3, 4, 5).
d. Response time in seconds.
e. Agent identity.
f. Determination identity (water, smoke, etc.).
g. Interference contaminant identity (exhaust gases, insecticides, smoke ingredient, etc.).

6.3.10 Field Operability

6.3.10.1 Air Tests

a. Record the following for each determination:

1) Test item identification number
2) Threshold chemical agent concentration (in applicable units)
3) Determination identity (air test)
4) Determination number (1, 2, 3, 4, 5)
5) Chemical agent identity
6) Response time, in seconds
7) Ambient temperature, in degrees F
8) Barometric pressure, in inches of Hg
9) Wind velocity in kilometers, per hour
10) Weather condition (clear, cloudy, fog, etc.)
11) Relative humidity, in percent
12) Detector kit identity

b. Record the number of false alarms
c. Retain all photographs

6.3.10.2 Water Tests

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a. Record the following for each determination:

1) Test item identification number
2) Threshold chemical agent concentration (in applicable units)
3) Determination identity (water test)
4) Determination number (1, 2, 3, 4, 5)
5) Chemical agent identity
6) Response time in seconds
7) Ambient temperature in degrees F
8) Barometric pressure in inches of Hg
9) Wind velocity in kilometers per hour
10) Weather condition (clear, cloudy, fog, etc.)
11) Relative humidity in percent
12) Detector kit identity

b. Record the number of false alarms
c. Retain all photographs

6.3.10.3 Surface Tests

a. Record the following for each determination:

1) Test item identification number
2) Threshold chemical agent concentration (in applicable units)
3) Determination identity (surface test)
4) Determination number (1, 2, 3, 4, 5)
5) Chemical agent identity
6) Response time in seconds
7) Surface plate identity
8) Ambient temperature in degrees F
9) Barometric pressure in inches of Hg
10) Weather condition (clear, cloudy, fog, etc.)
11) Wind velocity in kilometers per hour
12) Relative humidity in percent
13) Detector kit identity

b. Record the number of false alarms
c. Retain all photographs

6.3.11 Human Factors

Record the following for the test item, as applicable:

a. Carrying strap adequacy.
b. Ease of handling.
c. Ease of carrying.
d. Ease of operating.
e. Ease of interpretation of response (skill level required).
f. Ease of interpretation of responses under various lighting conditions.
g. Compatibility with other clothing and equipment.

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h. Ease of handling the test item when wearing protective clothing, gloves, etc. (including cold weather clothing).
i. Adequacy and simplicity of operating instructions.
j. Time required for interpreting response (too lengthy, etc.).
k. Factors which caused frequent complaints from operators.

6.4 DATA REDUCTIONS AND PRESENTATION

6.4.1 Receipt Inspection

a. Data collected as a result of this procedure shall be presented as indicated in the applicable portions of MTP 8-2-500.
b. The description of the item, number of items tested, and conditions upon receipt shall be presented in tabular form.
c. Photographs shall be used to substantiate results.

6.4.2 Safety Evaluation

a. A Safety Release Recommendation (USATECOM Regulation 385-6) shall be forwarded to U. S Army Test and Evaluation Command within 30 days of the beginning of the test. The Safety Release Recommendation shall contain the following information: Special safety considerations on hazards to personnel and materiel (including developmental types of equipment as well as standard components used in assemblage of items being tested).
b. Data and comments relative to safety hazards observed during any phase of testing.
c. Comments relative to suggested safety improvements.

6.4.3 Simulated Environmental Testing

a. The results of the subtests conducted shall be presented in tabular or other suitable form.
b. The results of the operational check tests performed at the conclusion of the various environmental tests shall be presented in narrative or other suitable form.

6.4.4 Rough Handling and Surface Transport

a. The results of this subtest shall be presented as indicated in applicable sections of MTP 8-2-503.
b. Present data on performance of test item after subjection to rough handling and surface transport tests in tabular form.
c. Narrative comments, photos, etc. shall be included, as required.

6.4.5 Air Transportability

a. The results of this subtest shall be presented as prescribed in MTP 7-2-515.
b. Air transport conditions shall be reported in tabular or other convenient form.
c. Narrative comments, photos, etc., may be included, if required.

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6.4.6 Air Drop Capability

6.4.6.1 Shipping Container and Field Use Tests

a. The results of these tests shall be presented as described in MTP 7-2-509.
b. Narrative comments, photos, etc. shall be included, as required.

6.4.7 Decontamination Aspects

a. Present the results as described in MTP 8-2-510.
b. Operability results shall be presented in tabular form.
c. Narrative comments shall be used to summarize results and the performance of the test items.

6.4.8 Maintenance Aspects

Data from this test shall be presented in narrative form showing what maintenance had been required. The report shall be supplemented by photos, drawings or other devices to substantiate the conclusions and recommendations.

6.4.9 Operational Characteristics

6.4.9.1 Sensitivity and Response

Present the following:

a. Data in tabular form to indicate:

1) Agents detected.
2) Methods of introducing aerosol or testing water or surfaces (include sampling results).
3) Responses by test item (include photographs of color indications as applicable) and make the necessary computations to show mean response time and threshold sensitivity.

b. Data in tabular form regarding responses of the test item under various conditions as compared to actual requirements or objectives.
c. Comments relative to deficiencies, shortcomings, or other inadequacies of the test item as regards operational sensitivity and response.
d. Photographs and narrative comments as necessary to substantiate conclusions and recommendations.

6.4.9.2 Interference Aspects

Present the following:

a. Data in tabular form to indicate:

1) Agents detected·

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2) Interference contaminant and method of introduction.
3) Medium contaminated (air, water, etc.).
4) Responses by the test item (include photographs of color indications, as applicable, and make the necessary computations to show mean response time and threshold sensitivity).

b. Present comparison tables to show the effects of interference.

c. Comments relative to deficiencies, shortcomings or other test item inadequacies.

d. Narrative comments, photos, etc. shall be included as necessary to substantiate conclusions and recommendations.

6.4.10 Field Operability

Present the following:

a. Data in tabular and photographic form to indicate the results of the various detection tests performed under field conditions and for comparison with "laboratory" results.

b. Compare the test item results with those for "standard" detection kits for the various tests performed.

c. Data regarding the type of agent used and the results of attempted detection relative to desired or required results.

d. Degree to which the test item satisfied governing criteria for performance.

e. When applicable, data regarding false responses and possible or probably causes.

f. Meteorological conditions shall be presented with the data to show their effect on test item performance.

g. Narrative comments, photos, etc. shall be used to supplement tabular data and to substantiate conclusions and recommendations.

6.4.11 Human Factors

a. Data from this subtest shall be presented in tabular, narrative, or other suitable form supplemented by photographs and graphic or art presentations as required.

b. A summary of comments regarding shortcomings and recommended improvements shall be presented.

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